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Introduction

The following notes were compiled to refresh my logic knowledge, based on The Logic Manual (Halbach, V., 2010)

Syntax and Semantics of Propositional Logic

The formula ψ is *logically true* iff $\emptyset \models \psi$, i.e. iff $\models \psi$. In propositional logic, a logically true formula is called a *tautology*.

Theorem 2.14 (Semantic entailment, Logical entailment, Logically implies, p. 43). For $n \geq 1$,

$$\{\psi_i\}_{i=1}^n \models \phi \iff \left(\bigwedge_{i=1}^n \psi_i \right) \rightarrow \phi \text{ is a tautology.}$$

or in other word:

$$\{\psi_i\}_{i=1}^n \models \phi \iff \models \left(\left(\bigwedge_{i=1}^n \psi_i \right) \rightarrow \phi \right).$$

Example. We can prove that a propositional formula is a tautology by enumerating every possible assignment of truth values to

$$((P \rightarrow \neg Q) \wedge Q) \rightarrow \neg P.$$

Its truth table is:

P	Q	$P \rightarrow \neg Q$	$(P \rightarrow \neg Q) \wedge Q$	$\neg P$	$((P \rightarrow \neg Q) \wedge Q) \rightarrow \neg P$
T	T	F	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	T	F	T	T

Since the final column contains only T, the formula is a tautology. Hence, by Theorem 2.14,

$$P \rightarrow \neg Q, Q \models \neg P.$$

Formalisation in Propositional Logic

L_1 is the formal language of propositional logic: sentence letters such as P , Q , and R ; connectives such as $\neg$, $\wedge$, $\vee$, $\rightarrow$, and $\leftrightarrow$; and formation rules determining which strings are sentences. For example, $\neg P$ and $(P \wedge Q)$ are sentences of L_1 .

A *propositional formula* is an expression built from sentence letters and connectives according to the formation rules. A *sentence* is a formula with no free variables. A sentence may be assigned a truth value relative to an interpretation or structure, e.g. a structure A might assign $A(P) = T$. (p. 87)

In propositional logic there are no free individual variables, so every well-formed propositional formula can be treated as a sentence. P , $\neg P$, and $P \wedge Q$ are therefore called sentences of L_1 , while ϕ and ψ are metavariables ranging over sentences of L_1 , e.g. ϕ may stand for $P \wedge Q$.

Connectives often include $\wedge$, $\vee$, and $\neg$, but a sentence-forming operator such as “it is necessarily the case that...” can also be a connective, although it is not part of classical propositional logic and is not truth-functional (see below)

[ADD THE ZENO EXAMPLE BELOW HERE]

A *direct subsentence* is the sentence “at the next level down to” the sentence in question, e.g. the direct subsentence of $\neg(P \wedge Q)$ is $P \wedge Q$, whereas the direct subsentences of $P \wedge Q$ are P and Q .

Characterisation 3.1 (Truth Functionality, p.54) A connective is *truth-functional* iff the truth-value of the compound sentence cannot be changed by replacing a direct subsentence with another sentence having the same truth-value.

The usual connectives $\wedge$, $\vee$, $\neg$, etc are always truth functional.

Example (Not truth functional). Let P be the sentence “ $2 + 2 = 4$,” and let Q be the sentence “London is the capital of the UK.” Both P and Q are true. However, under the usual interpretation of $\Box$ as “it is necessarily the case that,” $\Box P$ is true, whereas $\Box Q$ is false: London is the capital, but historically it could have turned out otherwise. Thus, replacing P with the equally true sentence Q changes the truth value of the compound sentence: although Q is true, $\Box Q$ is false because London is not *necessarily* the capital of the UK. Therefore, $\Box$ is not truth-functional.

Basics Recap. Recall $A \rightarrow B$ means “if A is true, then B follows as a consequence”. So, A is sufficient for B because A is enough to obtain B as a consequence. Whereas B is necessary for A because B follows from A .

Scope ambiguities can arise so we must use parentheses appropriately, e.g. $P \wedge (Q \vee R)$ versus $(P \wedge Q) \vee R$.

The Syntax of Predicate Logic

Consider

P : Zeno is a tortoise. Q : All tortoises are toothless. Therefore R : Zeno is toothless.

This argument is not propositionally valid since the two premisses and conclusion are formalised as three sentence letters P , Q and R respectively, but $P, Q \not\models R$.

Propositional logic treats each complete sentence as an atomic unit, i.e. there is no propositional connective linking the three sentence letters, and propositional logic does not look inside them to see the recurrence of Zeno, tortoise, and toothless.

We can represent formulae using e.g. Px or Pxy , sometimes we include the *arity-indices*, e.g. P^1x or P^2xy to indicate the number of variables accepted by P .

Example (p.89) Converting English sentences to predicate logic: “All frogs are amphibians” translates as “for all x (if x is a frog, then x is an amphibian)” or in symbols $\forall x(P(x) \rightarrow Q(x))$, with the predicate letters (not sentences) P : ... is a frog, and Q : ... is an amphibian. P and Q become formulae when supplied with a variable, e.g. $P(x)$, $Q(x)$. When a variable is bound, i.e. $\forall xP(x)$, $\exists xP(x)$, the formula becomes a sentence.

Example (p.92) “If it’s raining then Bill reads a book or a newspaper” translates as $P \rightarrow \exists x(P^2ax \wedge (Qx \vee Rx))$ with P : it’s raining, Q^1 : ... is a book, R^1 : ... is a newspaper, and P^2 : ... reads

Semantics of Predicate Logic

Meaning of “logically true”:

A sentence of the language L_2 is said to be *logically true* $\iff$ it is true under any interpretation of the constants and predicate letters.

Meaning of “valid”:

An argument in L_2 will be defined to be valid $\iff$ there is no interpretation under which the premisses are all true and the conclusion is false, i.e. $\Gamma \models \phi$ where Γ is the set of premisses and ϕ the conclusion.

Meaning of “structure”:

A *structure* $\mathcal{A}$ consists of a non-empty *domain* D and an interpretation function I . The domain D contains the objects under discussion, while I assigns:

- each constant a an element $I(a) \in D$;
- each unary predicate letter P a subset $I(P) \subseteq D$;
- each n -ary predicate letter R an n -ary relation $I(R) \subseteq D^n$.

For example, let $D = \mathbb{Z}$, and suppose a means zero, P means “... is even,” and R means “... is less than ...”. Then

$$\begin{aligned} I(a) &= 0, \\ I(P) &= \{n \in \mathbb{Z} : \exists k \in \mathbb{Z} (n = 2k)\}, \\ I(R) &= \{(m, n) \in \mathbb{Z}^2 : m < n\}. \end{aligned}$$

Consequently, $P(a)$ is true because $I(a) \in I(P)$. If $I(b) = 3$, then $R(a, b)$ is true because $(0, 3) \in I(R)$.

Semantic Value (see p. 99). Let $|e|_{\mathcal{A}}^{\alpha}$ be the *semantic value* of the expression e in the $\mathcal{L}_2$ -structure $\mathcal{A}$ under the variable assignment α (i.e. α is a map over $\mathcal{A}$). Let $\mathcal{A} = (D_{\mathcal{A}}, I_{\mathcal{A}})$, where $D_{\mathcal{A}}$ is the domain and $I_{\mathcal{A}}$ is the interpretation function that assigns semantic values to the non-logical symbols (i.e. constants and predicates):

- **Constant** t : $|t|_{\mathcal{A}}^{\alpha} \in D_{\mathcal{A}}$ assigned to t by $\mathcal{A}$, e.g. $|c|_{\mathcal{A}}^{\alpha} = 5$ if $c = 5$ by $\mathcal{A}$.
- **Variable** v : $|v|_{\mathcal{A}}^{\alpha} \in D_{\mathcal{A}}$ assigned to v by α , e.g. $|x|_{\mathcal{A}}^{\alpha} = 7$ if $\alpha(x) = 7$.
- **Sentence letter** P : $|P|_{\mathcal{A}}^{\alpha} \in \{T, F\}$ assigned directly by $\mathcal{A}$; α has no effect.
- **Formula / Sentence** ϕ : $|\phi|_{\mathcal{A}}^{\alpha} \in \{T, F\}$; α matters if ϕ has free variables. $|\phi|_{\mathcal{A}}^{\alpha} = T$ means “ α satisfies ϕ in $\mathcal{A}$ ”, sometimes written $\mathcal{A} \models \phi[\alpha]$. ϕ is a sentence if it has no free-variables.
- **Unary predicate** ϕ : $|\phi|_{\mathcal{A}}^{\alpha} \subseteq D_{\mathcal{A}}$, e.g. if E = “is even”, then $|E|_{\mathcal{A}}^{\alpha} = \{\dots, -2, 0, 2, 4, \dots\}$.
- **Binary predicate** ϕ : $|\phi|_{\mathcal{A}}^{\alpha} \subseteq D_{\mathcal{A}}^2$, e.g. if L = “<”, then $(2, 5) \in |L|_{\mathcal{A}}^{\alpha}$.

- **3-ary predicate** ϕ : $|\phi|_{\mathcal{A}}^{\alpha} \subseteq D_{\mathcal{A}}^3$, e.g. if $S = "x + y = z"$, then $(2, 3, 5) \in |S|_{\mathcal{A}}^{\alpha}$.
- **n -ary predicate** ϕ : $|\phi|_{\mathcal{A}}^{\alpha} \subseteq D_{\mathcal{A}}^n$, i.e. a set of ordered n -tuples, e.g. if $\phi(x_1, \dots, x_n) = "x_1 + \dots + x_n = a"$, then $(x_i)_{i=1}^n \in |\phi|_{\mathcal{A}}^{\alpha} \iff \sum_{i=1}^n x_i = a$.

Definition 5.2 (Satisfaction, pp.100-104):

- **Atomic (no connectives or quantifiers) formula**: $|\Phi t_1 \dots t_n|_{\mathcal{A}}^{\alpha} = T$ iff $(|t_1|_{\mathcal{A}}^{\alpha}, \dots, |t_n|_{\mathcal{A}}^{\alpha}) \in |\Phi|_{\mathcal{A}}^{\alpha}$, where Φ is an n -ary predicate letter, e.g. $Px_1x_2\dots x_n$, with $n \geq 1$, and each t_i is either a variable or constant.
- **Negation**: $|\neg\phi|_{\mathcal{A}}^{\alpha} = T$ iff $|\phi|_{\mathcal{A}}^{\alpha} = F$.
- **Conjunction**: $|\phi \wedge \psi|_{\mathcal{A}}^{\alpha} = T$ iff $|\phi|_{\mathcal{A}}^{\alpha} = T$ and $|\psi|_{\mathcal{A}}^{\alpha} = T$.
- **Disjunction**: $|\phi \vee \psi|_{\mathcal{A}}^{\alpha} = T$ iff $|\phi|_{\mathcal{A}}^{\alpha} = T$ or $|\psi|_{\mathcal{A}}^{\alpha} = T$.
- **Conditional**: $|\phi \rightarrow \psi|_{\mathcal{A}}^{\alpha} = T$ iff $|\phi|_{\mathcal{A}}^{\alpha} = F$ or $|\psi|_{\mathcal{A}}^{\alpha} = T$, the reason for which can be seen directly from the truth table for $\rightarrow$: TTT , TFF , FTT , FFT .
- **Biconditional**: $|\phi \leftrightarrow \psi|_{\mathcal{A}}^{\alpha} = T$ iff $|\phi|_{\mathcal{A}}^{\alpha} = |\psi|_{\mathcal{A}}^{\alpha}$.
- **Existential quantifier**: $|\exists v\phi|_{\mathcal{A}}^{\alpha} = T$ iff there is some value in $D_{\mathcal{A}}$ that can be assigned to v , while leaving the values of all other variables fixed by α , for which ϕ is true. Formally, $|\exists v\phi|_{\mathcal{A}}^{\alpha} = T \iff |\phi|_{\mathcal{A}}^{\beta} = T$ for some assignment β such that $\beta(u) = \alpha(u)$ for every variable $u \neq v$.
- **Universal quantifier**: $|\forall v\phi|_{\mathcal{A}}^{\alpha} = T$ iff ϕ is true for every possible value in $D_{\mathcal{A}}$ assigned to v , while leaving the values of all other variables fixed by α . Formally, $|\forall v\phi|_{\mathcal{A}}^{\alpha} = T \iff |\phi|_{\mathcal{A}}^{\beta} = T$ for every assignment β such that $\beta(u) = \alpha(u)$ for every variable $u \neq v$.

[TODO: **Examples**: show the examples i read through on pp.105-108, i understood them all.]

Definition 5.7 (Logical truth, contradiction, equivalence & semantic consistency, p.108):

- **Logically true**: A sentence ϕ is logically true iff ϕ is true in every $\mathcal{L}_2$ -structure $\mathcal{A}$; i.e. $\mathcal{A} \models \phi$ for every $\mathcal{A}$.
- **Contradiction**: A sentence ϕ is a contradiction iff ϕ is false in every $\mathcal{L}_2$ -structure $\mathcal{A}$; i.e. $\mathcal{A} \not\models \phi$ for every $\mathcal{A}$.
- **Logically equivalent**: Sentences ϕ and ψ are logically equivalent iff they are true in exactly the same $\mathcal{L}_2$ -structures; i.e. $\mathcal{A} \models \phi \iff \mathcal{A} \models \psi$ for every $\mathcal{A}$.
- **Semantically consistent**: A set Γ of $\mathcal{L}_2$ -sentences is *semantically consistent* iff there is at least one $\mathcal{L}_2$ -structure $\mathcal{A}$ in which every sentence in Γ is true; i.e. there exists some $\mathcal{A}$ such that $\mathcal{A} \models \phi$ for every $\phi \in \Gamma$. Γ is *semantically inconsistent* iff no such $\mathcal{A}$ exists.

Definition 5.8 (Validity; Valid Argument, p.109): Let Γ be a set of $\mathcal{L}_2$ -sentences (premises) and ϕ an $\mathcal{L}_2$ -sentence (conclusion). The argument is *valid* iff there is no $\mathcal{L}_2$ -structure $\mathcal{A}$ in which every sentence in Γ is true and ϕ is false. Equivalently, every $\mathcal{L}_2$ -structure that makes all the premises true also makes the conclusion true. This is written $\Gamma \models \phi$ and read “ Γ logically implies ϕ ” or “ ϕ logically follows from Γ .” We also say “semantically entails” or “logically entails” for “logically implies”. **In other words:** There is no structure $\mathcal{A}$ in which all the premises in Γ are true but the conclusion ϕ is false.

Note: This is closely related to *tautology*, but that term is usually reserved for propositional logic. In predicate logic we instead say a sentence is *logically true* iff it is true in every structure.

An $\mathcal{L}_2$ structure $\mathcal{A}$ is a *counterexample* to an argument iff all premises of the argument are true in $\mathcal{A}$ and the conclusion is false in $\mathcal{A}$.

[TODO: work through the counterexample examples]

Natural Deduction

Note: the following notes use the **bussproof** which works with Obsidian, but this is easily adapted to **ebproof** or plain LaTeX instead.

Basics.

- As example, $P \wedge Q, P \rightarrow R \vdash P \vee Q$ reads as: the premises (left) *prove/derive* ($\vdash$) the conclusion $P \vee Q$, which is different from *logical implication* ($\models$) - see below.
- The premises $P \wedge Q$ and $P \rightarrow R$ are free to use throughout the deduction.
- As we proceed we may introduce temporary assumptions, e.g. P , but when an assumption is discharged by a rule we enclose it in square brackets $[P]$ to show it has been discharged.
- Various rules are applied to deduce conclusions from premises, and so the proof tree builds until we reach the desired final conclusion. The rules are:
 - $\wedge$ Intro (*introduction*), $\wedge$ Elim (*elimination*)
 - $\vee$ Intro₁ (*left*), $\vee$ Intro₂ (*right*), $\vee$ Elim
 - $\rightarrow$ Intro, $\rightarrow$ Elim
 - $\neg$ Intro, $\neg$ Elim

Theorem (Soundness). $\Gamma \vdash \varphi \Rightarrow \Gamma \models \varphi$.

If φ is provable from Γ , then φ is logically implied by Γ .

Theorem (Completeness). $\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi$.

If φ is logically implied by Γ , then φ is provable from Γ .

Theorem (Adequacy). $\Gamma \models \varphi \iff \Gamma \vdash \varphi$.

$\Gamma \vdash \varphi$ means: there is a formal proof of φ from Γ using the allowed rules of deduction. Whereas $\Gamma \models \varphi$ means: in every interpretation where all sentences in Γ are true, φ is also true.

For example $P, P \rightarrow Q \vdash Q$ because Q can be formally derived by modus ponens, and $P, P \rightarrow Q \models Q$ because there is no interpretation in which both premises are true and Q is false.

So $\vdash$ is about proofs, whereas $\models$ is about truth in all models, i.e. logical consequence; when there are no premises, $\emptyset \models \varphi$, or simply $\models \varphi$, expresses logical validity, analogous to a *tautology* in propositional logic.

Example 6.3 $P \vee Q, P \rightarrow R \vdash R \vee Q$. **Proof.**

$$\frac{\frac{P \vee Q}{\frac{[P] \quad P \rightarrow R}{R} (\rightarrow \text{Elim})} \quad \frac{[Q] \quad P \rightarrow R}{R \vee Q} (\vee \text{Intro}_1)}{R \vee Q} (\vee \text{Elim})$$

Explanation. The premises are $P \vee Q$ and $P \rightarrow R$. Since we know $P \vee Q$ we can assume P in one branch and Q in another and see if we can derive the conclusion $R \vee Q$ from each. With P we know R since $P \rightarrow R$. Now we can trivially introduce $R \vee Q$. On the other hand, if we assume Q then we can trivially deduce $R \vee Q$. Finally, since $P \vee Q$, we have covered both branches and therefore $R \vee Q$.

Example 6.4 $\neg(P \rightarrow Q) \vdash \neg Q$. **Proof.**

$$\frac{\frac{[Q] \quad \neg(P \rightarrow Q)}{P \rightarrow Q} (\rightarrow \text{Intro})}{\neg Q} (\neg \text{Intro})$$

Explanation. To prove a negative such as $\neg Q$ we can use proof by contradiction. First assume Q , then by using $(\rightarrow \text{Intro})$ we can introduce $P \rightarrow Q$. On the other hand, we already have the premise $\neg(P \rightarrow Q)$. Given we have reached both ϕ and $\neg\phi$ we conclude in fact $\neg Q$, i.e. our original assumption Q must be false. **Note.** here we introduce a “ $\neg$ ” hence $(\neg \text{Intro})$, which contrasts with the next example of $(\neg \text{Elim})$ to remove the “ $\neg$ ”.

Example 6.5 $\neg P \rightarrow Q, \neg Q \vdash P$. **Proof.**

$$\frac{\frac{[\neg P] \quad \neg P \rightarrow Q}{Q} (\rightarrow \text{Elim})}{P} (\neg \text{Elim})$$

Explanation. Essentially this is just another proof by contradiction. We assume $\neg P$ and use the premise $\neg P \rightarrow Q$ which by $(\rightarrow \text{Elim})$ gives Q , which together with $\neg Q$ forms a contradiction, so we may invoke $(\neg \text{Elim})$ to discharge $\neg P$, and conclude P . Here, instead of introducing a negation as in Example 6.4, $(\neg \text{Elim})$ discharges the negated assumption $\neg P$ and concludes the corresponding unnegated sentence P .

Example 6.6 $\vdash P \vee \neg P$. **Proof.**

$$\frac{\frac{\frac{[P]}{P \vee \neg P} (\vee \text{Intro}_1) \quad [\neg(P \vee \neg P)]}{\neg P} (\neg \text{Intro})}{\frac{\frac{\neg P}{P \vee \neg P} (\vee \text{Intro}_2) \quad [\neg(P \vee \neg P)]}{P \vee \neg P} (\neg \text{Elim})}$$

Explanation. From the empty set $\emptyset$ we assume P , for we have no premises to use, and introduce $P \vee \neg P$. Next, assume $\neg(P \vee \neg P)$ to reach a contradiction, leading to the discharge of our first assumption P , i.e. $[P]$, and conclude the interim step $\neg P$. We then introduce $P \vee \neg P$ and again assume $\neg(P \vee \neg P)$, another contradiction through which negation elimination discharges the two $\neg(P \vee \neg P)$ terms to conclude $P \vee \neg P$, as required.

Example 6.7 $\vdash (P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$. **Proof.**

$$\frac{\frac{\frac{[P \rightarrow Q] \quad [P]}{Q} \quad [\neg Q]}{\neg P} (\neg \text{Intro}) \quad \frac{\frac{[\neg Q \rightarrow \neg P] \quad [\neg Q]}{\neg P} \quad [P]}{Q} (\neg \text{Elim})}{\frac{\frac{\neg P}{\neg Q \rightarrow \neg P} (\rightarrow \text{Intro}) \quad \frac{Q}{P \rightarrow Q} (\rightarrow \text{Intro})}{(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)} (\leftrightarrow \text{Intro})}$$

Explanation. We have no premises with which to start, so (branch 1) first assume $P \rightarrow Q$ and P , to obtain Q and assume $\neg Q$ to form a contradiction to introduce $\neg P$ which discharges P . Next, since $\neg Q$ up until now, we may introduce $\neg Q \rightarrow \neg P$. On the other hand, (branch 2) assume $\neg Q \rightarrow \neg P$ and $\neg Q$ to obtain $\neg P$, then assume P to reach a contradiction and eliminate to obtain Q , and discharge $\neg Q$, and then introduce $P \rightarrow Q$. Now, branch 1 gives $(P \rightarrow Q) \rightarrow (\neg Q \rightarrow \neg P)$ and branch 2 gives $(\neg Q \rightarrow \neg P) \rightarrow (P \rightarrow Q)$, hence we introduce (conclude) that $(P \rightarrow Q) \leftrightarrow (\neg Q \rightarrow \neg P)$, as required, which also discharges the initial branch assumptions $P \rightarrow Q$ and $\neg Q \rightarrow \neg P$.

Example 6.8 $\neg(Q \wedge \neg R) \vdash Q \rightarrow R$. **Proof.**

$$\frac{\neg(Q \wedge \neg R) \quad \frac{\frac{[Q] \quad [\neg R]}{Q \wedge \neg R} (\wedge \text{Intro})}{\neg R} (\neg \text{Elim})}{\frac{R}{Q \rightarrow R} (\rightarrow \text{Intro})}$$

Explanation. Assume Q and $\neg R$ and introduce $Q \wedge \neg R$, then use the premise to reach a contradiction and eliminate $\neg R$ to obtain R . Finally introduce $Q \rightarrow R$ which discharges Q .

Example 6.9 $P, \neg P \vdash Q$. **Proof.**

$$\frac{P \quad \neg P}{Q}$$

Explanation. Halbach's definition of $(\neg \text{Elim})$ is "The result of appending a sentence ϕ to a proof of ψ and a proof of $\neg\psi$ and of discharging **all** assumptions of $\neg\phi$ in both proofs is a proof of ϕ ." Note the word "all" could be replaced with "zero or more". Since there are no assumptions on $\neg Q$ then "all" here relates to zero cases, and as such the rule still applies, hence Q .

TODO - continue to type up my book notes when I get time...